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Problem Chosen

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Summary Sheet

Owing to the increased accumulation of CO₂ in the atmosphere, the impacts of climate change on people and nature are increasingly apparent. Showing our concerns about this serious global issue and tending to provide practical solutions, we offer unique insight through the construction of mathematical models. Specifically, our team builds three distinctive models, including the regression model and autoregressive integrated moving average model (ARIMA), and back propagation neural network (BP), which are aimed at characterizing the past, and making predictions about the future CO₂ concentration level as well as future global annual mean surface-air temperature (GMSAT).

To fit the provided CO₂ concentration level and GMSAT dataset, we apply all three models. By evaluating R^2 (goodness-of-fit), we find that the ARIMA model has the best capacity in describing the past CO₂ data ($R^2 = 0.999$). It also reveals that the CO₂ concentration level will hit 497.97ppm in 2050 and 770.57ppm in 2100. (OECD predicts a CO₂ level of 685 in 2050). Furthermore, both the quadratic regression model and the ARIMA model can be deemed as the most appropriate model to predict future GMSAT. In particular, in 2030, GMSAT is likely to be 1.25C above the temperature mean of the base period 1951-1980 and it will continue rising, being 1.5C and 2C above the temperature mean in 2038 and 2052 respectively.

Moreover, we apply the Pearson correlation coefficient (PCC) and Spearman's rank correlation coefficient (SRCC) and ultimately determine that there is a strong linear and monotonically increasing relationship between CO₂ concentration and land-ocean temperature. To be more precise, with the theoretical base of the linear regression model and ARIMA, we construct a correlation model between CO₂ and temperature, finding that a doubling of the amount of carbon dioxide in the atmosphere would cause global temperature to rise by between 4 and 4.5 degrees Celsius. In the end, we discover that our correlation model has a high reliability of about 20 years based on the anticipated CO₂ and temperature data.

Beyond basic theoretical models, we comprise our innovations. To illustrate, we use the ARIMA model to optimize the linear regression model and thus raise R^2 of the correlation model. Meanwhile, within BP and ARIMA models, we encompass sensitivity analysis to make the explanation as comprehensive as possible. We also write a non-technical article to interpret the results outputting by model and present our concerns about warming global temperature, demonstrating our plausible suggestions.

As for our strengths, the future CO₂ concentration and GMSAT are predicted by multiple models, each of which has its advantage and feature. Hardly can we ignore the high value of R^2 , since it makes the prediction of GMSAT and CO₂ concentration accurate. By contrast, our future expectation focus on optimizing the BP model by collecting more precise data. Another addition is that we tend to account for more variables in our regression model, creating multivariate regression to face the upcoming climate issues and make more precise forecasts.

Keywords: Global Warming; CO₂ concentration; Regression; ARIMA; BP; Correlation

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1 Introduction

1.1 Background

Climate change can be deemed as an issue that trigger heated discussions in human living, which led to the massive insights of the scientists, policymakers, even our senior students. Among various extreme weather phenomena, elevated temperatures account for the largest attention. In other words, our earth is experiencing global warming, which contributes to hazardous consequences, including the glacier disintegration, destructing of food chains, increasing of climate extreme events, and etc. More seriously, it is no exaggeration to say that increasing global temperature is threatening the survival of all creatures on earth, including human beings.

The intensification of global warming can be attributed to several distinctive aspects, and some scientists assert that the rise of carbon dioxide (CO₂) concentration, as a significant Greenhouse gas, that cannot be ignored. Based on the data provided by National Oceanographic and Atmospheric Administration (NOAA) and Scripps Institution of Oceanography (SIO), until 2021, the global average atmospheric carbon dioxide concentration was 416.45 parts per million, setting a new record high despite the continued global economic drag from the COVID-19 pandemic.

Under such context, accurate prediction of future CO₂ concentration levels and temperature as well as finding their relationship is a crucial task. Specifically, in the face of threats of global warming, accurate predictions will increase individual awareness of the seriousness of the climate changes and challenges, allow scientists to assess more thoroughly and effectively on the future changes of the earth climate system, and help to maintain the sustainable development of the earth, the place where all humans reside.

1.2 Problem Restatement

- **Question 1:** To begin with, based on the given CO₂ concentration data, we need to find out the largest increase in CO₂ levels over time interval from 1994 to 2004 and to construct a variety of mathematical models to analyze the past and predict future CO₂ concentration levels, identifying the consistency between the model we built and the expectational data given by NOAA, SIO, and OECD. We will solve Q1 in section 3,4,5,and 6.
- **Question 2:** Following that, the global annual mean surface-air temperature (GMSAT) data over the past 64-year are also required to be explored, and it is also needed to predict future change based on the analysis. Furthermore, we also need to recognize and justify if there is any potential correlation between CO₂ concentration levels and land-ocean temperature, estimating the applicability of the model in the future and elaborating on our concerns. We will solve Q2 in section 7.
- **Question 3:** In this question, we write a non-technical journal to present our results. Based on a variety of mathematical models built by ourselves, we predicted the future CO₂ concentration level and temperature in 2050 and 2100 respectively. We also elaborate the relationship between CO₂ and temperature and make feasible recommendations for the future.

2 Preparation

2.1 Assumption and Justification

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

- **Assumption 1:** There will be no climate factors happening in the future that significantly affects the routine change of CO₂ concentration level and GMSAT.
- **Justification 1:** Predicting regular changes in CO₂ concentration and temperature in the presence of extremely unstable factors is literally impossible. To simplify the problem and make our prediction reliable, we will assume the nonexistence of such factors.
- **Assumption 2:** We assume the ecosystem of earth could maintain stable in the future.
- **Justification 2:** We predict the future CO₂ concentration level and GMSAT based on historical data, which must rely on a stable ecosystem, otherwise climate will be extremely affected and we cannot get accurate forecasts.
- **Assumption 3:** Human technology will not show significant progress, and human beings will still mainly use existing energy sources
- **Justification 3:** We need to eliminate any external factors that could potentially affect current CO₂ concentration level and GMSAT. We assume there will be no technological improvement, so that CO₂ emission and human influence on temperature can maintain a steady trend.

2.2 Visualization of Observed CO₂ and Temperature Data

In this section, we will provide the visualization of the observed of CO₂ concentration level (ppm) and GMSAT (°C) data.

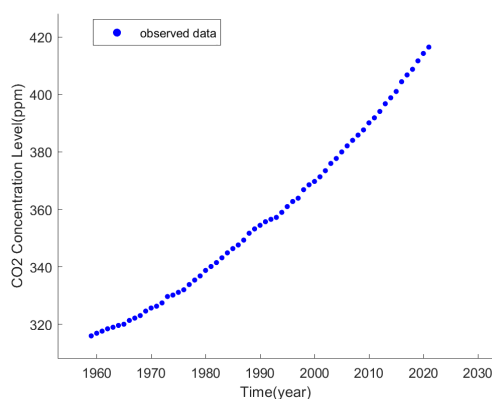


Figure 2.1: Scatter Diagram of CO₂

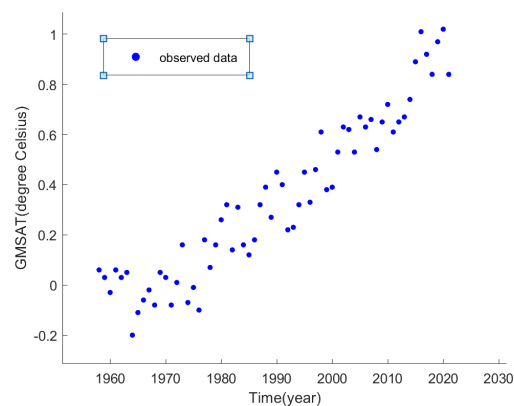


Figure 2.2: Scatter Diagram of Temp

2.3 Variables

Table 2.1: Variables Table

Variables	Description
C	CO2 Concentration Level (ppm)
T	Land-ocean temperature (degree celsius)
R	Residual Sums of Squares
ACF	Auto Correlation Function
$PACF$	Partial Auto Correlation Function
ε	Noises
R^2	Goodness-of-fit
R_p	Pearson Correlation Coefficient
R_s	Spearson's Rank Correlation Coefficient
x_t	Time (year)
$x_i (i = 1, 2, \dots, n)$	Input of Neural Network
$y_j (j = 1, 2, \dots, m)$	Hidden Layer Output(Also the Input of the Output Layer)
$o_k (k = 1, 2, \dots, m)$	Output of Neural Network
a	Threshold of Neurons In the Hidden Layer
b	Threshold of Neurons In the Output Layer
v_{ij}	Weight Matrix Between the Input and the Hidden Layer
w_{ij}	Weight Matrix Between the Hidden and the Output Layer

3 The Regression Model of CO2 and Temperature

3.1 The Least Squares Method (LSM)

The Least Squares Method (LSM) is the fundamental theory that guides our building of the regression model. Using LSM allows us to find the mathematical expression as a curve or line that best fits based on the given dataset.

We have a set of data $\{(x_{ti}, y_i) | i = 1, 2, \dots, n\}$, in which y_i can be C_i or T_i . The function that we apply to fit in the dataset is called $f(x)$. We first use Residual Sums of Squares (RSS) to evaluate the discrepancy between the actual value of CO2 or temperature and the predicted value provided by our hypothetical function:

$$R = \sum_{i=1}^N [y_i - f(x_{ti})]^2 \quad (3.1)$$

where, R represents RSS, $f(x_{ti})$ indicates the CO2 concentration level or GMSAT prediction value of the function at a specific time x_{ti} , and x_{ti} represents time which varied from 1959 to 2021. y_i is the observed CO2 or temperature data at the same time. In our regression model, we will assume our hypothetical function $f(x_{ti})$ to be linear function and non-linear function respectively.

In order for our hypothetical function to better fit the given dataset, we want to reduce the difference between the actual CO2 or temperature value and the predicted value. Therefore, we must determine the minimum RRS:

$$\min R = \min \sum_{i=1}^N [y_i - f(x_{ti})]^2 \quad (3.2)$$

3.2 The Linear Regression Model of CO2 and Temp

In our first scenario, we assume that the hypothetical function $f(x_t)$ is linear:

$$f(x_t) = ax_t + b \quad (3.3)$$

As we mentioned above, to find the best fit slope a and y-intercept b , we must determine the minimum RSS as:

$$\min R = \min \sum_{i=1}^N (y_i - ax_{ti} - b)^2 \quad (3.4)$$

We can find the minimum RSS by taking the partial derivatives and making each partial derivative equal to 0, hence equation (3.4) can be simplified into:

$$\begin{cases} \frac{\partial R}{\partial a} = \sum_{i=1}^N 2(y_i - ax_{ti} - b) \cdot (-x_{ti}) = 0 \\ \frac{\partial R}{\partial b} = \sum_{i=1}^N 2(y_i - ax_{ti} - b) \cdot (-1) = 0 \end{cases} \quad (3.5)$$

The two equations can be further simplified and we can find the value of a and b by the system of equation shown below:

$$\begin{cases} (\sum_{i=1}^N x_{ti}^2)a + (\sum_{i=1}^N x_{ti})b = \sum_{i=1}^N x_{ti}y_i \\ (\sum_{i=1}^N x_{ti})a + (\sum_{i=1}^N 1)b = \sum_{i=1}^N y_i \end{cases} \quad (3.6)$$

3.3 The Non-linear Regression Model of CO2 concentration and GMSAT

In the second scenario, we assume that the hypothetical function $f(x_t)$ is polynomial, which can be represented by:

$$f(x_t) = w_0 + w_1x_t + w_2x_t^2 + \dots + w_mx_t^m = \sum_{j=0}^m w_jx_t^j \quad (3.7)$$

In this case, we must determine the degree of polynomial m before regression. For instance, $j = 1$ stands for linear regression and $j = 2$ stands for quadratic regression.

When the degree of polynomial is determined, the minimum RSS becomes:

$$\min R = \min \sum_{i=1}^N (y_i - \sum_{j=0}^m w_jx_i^j)^2 \quad (3.8)$$

We can find the minimum value of RSS by taking the m partial derivatives and making each partial derivative 0.

One of the terms $\frac{\partial R}{\partial w_j} = 0, j = 1, 2 \dots m$ can be expanded and written as :

$$\frac{\partial R}{\partial w_j} = \sum_{i=1}^N 2x_{ti}^{j-1} (y_i - \sum_{j=1}^m w_j x_{ti}^j) = 0 \quad (3.9)$$

It can be shown to be a monotonically increasing primary function with respect to w_j , so the value we get after 1st partial derivative can be shown to be minimal.

The remaining terms in the system of equations mentioned above can also be expanded into an equation that contains the fitting coefficients. Thus, the m equations in the system of equations can be solved for m coefficients.

4 The Autoregressive Integrated Moving Average Model (ARIMA)

In this section, we will construct an autoregressive model, which is frequently used to represent time-varying nature phenomena. Consequently, it is excellent for characterizing the past and making predictions about the future CO2 concentration and land-ocean temperature.

4.1 Selection of the time-series models

The prerequisite for most autoregressive models is a stable initial dataset (stable in this case refers to the data mean, variance and other properties do not change with respect to the change of time, so that the data does not have obvious trends and periodic changes).

Therefore, at the beginning of our model's construction, we use the auto correlation function (ACF) to distinguish whether our initial data is statistically stable. ACF is used to describe the CO2 concentration level or GMSAT data's fluctuation over time, and it is a measurement to evaluate the impact of historical data on the present.

For time-series $\{(y_t) | t = 1, 2, \dots, T\}$ (y_t is CO2 concentration level or observed GMSAT), the ACF can be expressed as:

$$ACF = \sum_{t=h+1}^T \frac{(y_t - \bar{y})(y_{t-h} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2} \quad (4.1)$$

In the equation given above, the correlation coefficient of y_t and y_{t-h} is called the ACF of the y_t for interval h .

By analyzing the initial time series data, we obtained two ACF diagrams for the two variables CO2 concentration level and land-ocean temperature. (Figure 4.1 and Figure 4.2 respectively).

All of the stable time-series ACF images show a propensity for the images to drop quickly to zero and vary erratically in the vicinity of it. (significant at lag q , cut off after lag q). Based on the ACF diagrams, however, the original time-series CO2 and land-ocean temperature data display geometric decays (tails off), so for both CO2 concentration and land-ocean temperature is not statistically stable.

4.2 Determination of Parameters

Therefore, due to the instability of the original CO2 and land-ocean temperature data, we choose to build a Autoregressive Integrated Moving Average Model (ARIMA).

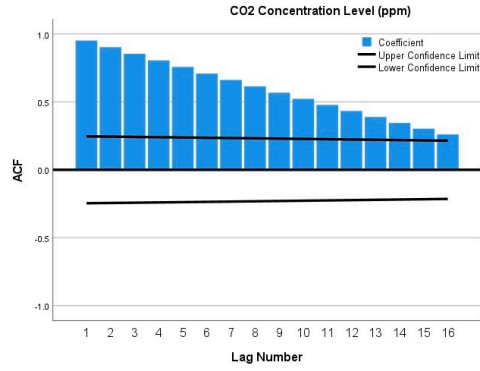


Figure 4.1: ACF of CO2

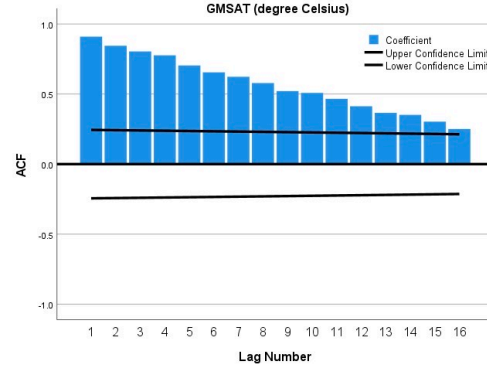


Figure 4.2: ACF of Temp

ARIMA is a integrated model combining 2 time-series models, AR and MA, and it also can be used to deal with statistically unstable time series. The expression for ARIMA model is shown below:

$$y'_t = \phi_1 y'_{t-1} + \dots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t = \sum_{i=1}^p \phi_i y'_{t-i} + \sum_{i=1}^q \theta_i \varepsilon_{t-i} + \varepsilon_t \quad (4.2)$$

where $y'(\cdot)$ is the stabilized time-series data, $\sum_{i=1}^p \phi_i y'_{t-i}$ is from AR model, and $\sum_{i=1}^q \theta_i \varepsilon_{t-i}$ is from MA model. ε_t is a random variable called noise.

Hence, to construct our AMIMA model, we must determine 3 parameters- p,d,q, which we will expound in the following 3 small sections.

4.2.1 Determination of d

Usually we use differencing to stabilize the time series by removing trend and seasonality. d denotes the time of differences needed to stabilize the initial time series data. We want to difference until the geometric decay is eliminated.

When d=2, we find that CO2 time series is stabilized , and when d=1, GMSAT time series also achieves stationarity .

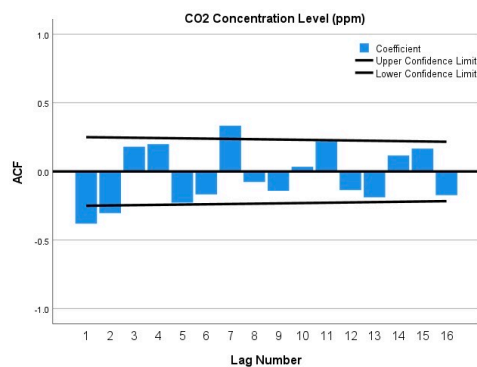


Figure 4.3: ACF of CO2 (d=2)

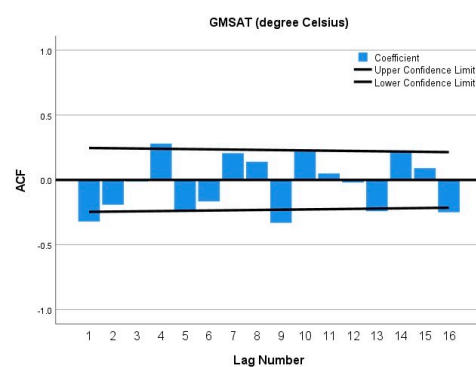


Figure 4.4: ACF of GMSAT (d=1)

4.2.2 Determination of p

p is the number of autoregressive terms. Usually we determine p by analysing the partial auto correlation function (PACF), which refers to the auto correlation coefficient between the variables after removing the indirect effects.

$$PACF = \frac{E[(y_t - \hat{E}(y_t))(y_{t-k} - \hat{E}(y_{t-k}))]}{E[(y_{t-k} - \hat{E}(y_{t-k}))^2]} \quad (4.3)$$

where $\hat{E}(x_t) = E[x_t | x_{t-1}, \dots, x_{t-k+1}]$, and $\hat{E}(x_{t-k}) = E[x_{t-k} | x_{t-k}, \dots, x_{t-k+1}]$. $\hat{E}$ represents all possible values of a discrete random variable.

We can determine p by observing the PACF diagram. To be more specific, p is the position where the graph cuts off after it.

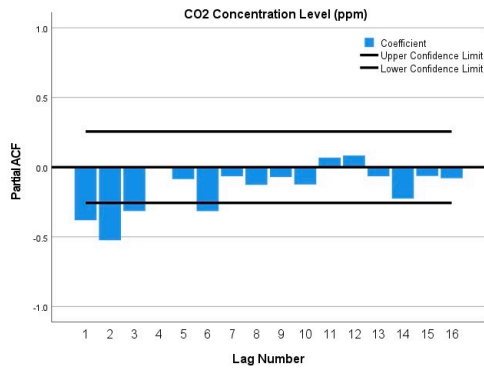


Figure 4.5: PACF of CO2 (d=2)

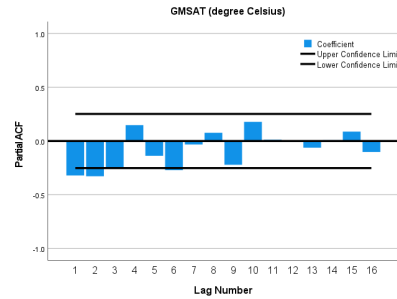


Figure 4.6: PACF of GMSAT (d=1)

So, based on Figure 4.5 and Figure 4.6 we can determine the value of P. In both CO2 and temperature PACF ARIMA model, p=3.

4.2.3 Determination of q

q is the number of lagged forecast errors in the prediction equation. The approach described in 3.2.2 can also be used to calculate the value of q, although in this instance, an ACF diagram is needed to be analyzed rather than a PACF.

In accordance with Figure 4.3 and Figure 4.4, in both CO2 and GMSAT ARIMA model, q=2.

4.3 Result

Based on the results of 4.2.1, 4.2.2 and 4.2.3, we need to build an ARIMA(3,2,2) model for CO2 and an ARIMA(3,2,1) model for temperature:

$$C_t = ARIMA(3, 2, 2) = \sum_{i=1}^3 \phi_i C'_{t-i} + \sum_{i=1}^2 \theta_i \varepsilon_{t-1} + \varepsilon_t \quad (4.4)$$

$$T_t = ARIMA(3, 1, 2) = \sum_{i=1}^3 \phi_i T'_{t-i} + \sum_{i=1}^2 \theta_i \varepsilon_{t-1} + \varepsilon_t \quad (4.5)$$

4.4 Extended Regression Model

Recap our memory, in our previous chapter, we talk about the basic linear and non-linear regression model as well as how we use LSM to determine the best fit coefficients.

To enhance the prediction accuracy of the basic regression model, here we continue utilizing the ARIMA model to optimize the function. Because natural climate change has noise and fluctuations, we also want such uncertainty to be considered.

To achieve that, we fit the residual term of the regression model again using the ARIMA model and construct a multivariate linear regression model by considering $f(x)$ and $ARIMA(p,d,q)$ as two variables, the expression of multivariate linear regression for 2 variables is given below:

$$y = \beta_0 + \beta_1 f(x) + \beta_2 ARIMA(q, d, q) + \varepsilon \quad (4.6)$$

In Equation (4.6) given above, this functions stand for the extended models for our regression models.

5 Back Propagation Neural Network (BP)

5.1 General Theory

In this section, aiming at a more accurate prediction of future values of CO2 and temperature, we build our prediction models based on neural network.

Among many neural networks, we choose BP neural network algorithm as the base modeling because BP has been relatively mature both in terms of network theory and performance. Its most prominent advantage is its strong nonlinear regression capability, which is advantageous in predicting future trends of CO2 and temperature.

For the prediction of CO2 concentration and temperature, we choose to build a traditional 3-layer BP neural network model (as shown in Figure 5.1, the 3 layers are input layer, hidden layer and output layer respectively).

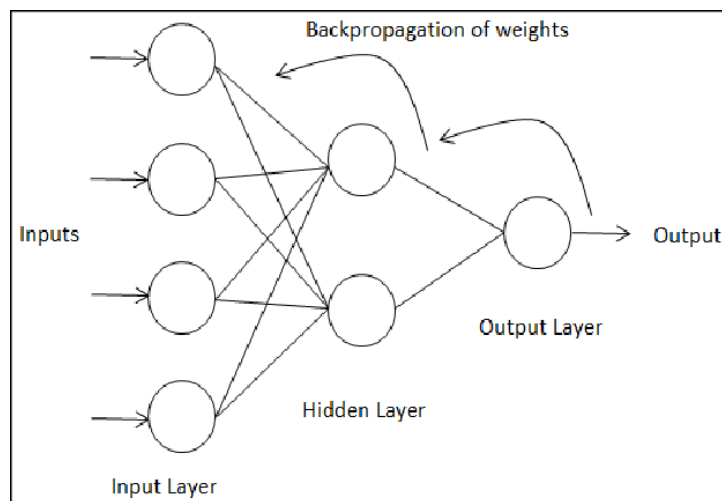


Figure 5.1: General Structure of BP Network [1]

5.2 Construction of BP model

The following procedure is our construction of the BP neural network model:

1. Initialize the network. We need to determine the number of neurons in the input and output layers of BP network based on the input $X = (\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n)$ and the desired output $D = (\vec{d}_1, \vec{d}_2, \dots, \vec{d}_l)$. We also initialize the connection weights and thresholds between neurons in each layer and render the learning rate and transfer function.

2. The number of neurons is also need to be determined, we usually use the following equation in a 3-layer BP network:

$$N_{hide} = 2N_{in} + 1 \quad (5.1)$$

The equation given above is based on the Kolmogorov-Arnold representation theorem [2]. where N_{hide} is the number of neurons in the hidden layer and N_{in} is the number of neurons in the input layer, which we already determined in step 1.

3. The next step is to calculate the output of the hidden layer, which is given by:

$$y_i = f\left(\sum_{i=1}^n v_{ij}x_i - a_j\right) = f\left(\sum_{i=0}^n v_{ij}x_i\right) \quad (5.2)$$

where $v_{0j} = -1$, $x_0 = a_j$, $f(\cdot)$ is the transfer function of the hidden layer.

4. After we determined the output of the hidden layer, we need to calculate the output of the output layer.

$$o_k = \varphi\left(\sum_{j=1}^m w_{jk}y_j - b_k\right) = \varphi\left(\sum_{j=0}^m w_{jk}y_j\right) \quad (5.3)$$

In equation 5.3, $w_{0k} = -1$, $y_0 = b_k$, and $\varphi(\cdot)$ is the transfer function of the output layer.

5. We calculate the total error of the network.

$$EN = \frac{1}{2}(D - O)^2 = \frac{1}{2} \sum_{k=1}^l (d_k - o_k)^2 \quad (5.4)$$

6. Update weight matrix. Based on the total error of the network, we update the weight matrix by the following formula:

$$\begin{aligned} v_{ij} &= v_{ij} + \Delta v_{ij} \\ w_{jk} &= w_{jk} + \Delta w_{jk} \\ \delta v_{ij} &= \eta \left(\sum_{k=1}^1 \delta_k^0 w_{jk} \right) y_j (1 - y_j) x_i \\ \Delta w_{jk} &= \delta \eta_k^0 y_j \end{aligned} \quad (5.5)$$

where $\eta_k^0 = (d_k - o_k)o_k(1 - o_k)$, η stands for learning rate.

7. Determine whether the overall network error meets the accuracy requirement, if not, we will let the network return to step 2.

6 Results for Regression Model, ARIMA and BP Model

In this section, we will primary focus on predicting the future CO2 concentration level and GMSAT and compare the predictive precision among all 3 distinctive models. The main indicator that we will utilize is called R^2 .

R^2 is the coefficient of determination, which is always used to assess the goodness-of-fit. It represents the percentage of variation in the dependent variable that can be predicted from the independent variable. The goodness-of-fit is better when R^2 is closer to 1. R^2 is represented by the following expression:

$$R^2 = 1 - \frac{RSS}{TSS} \quad (6.1)$$

where $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$ and $RSS = \sum_{i=1}^n (y_i - f(x_t))^2$, which we mentioned in 3.2. We will consider $R^2 = 9$ to be an appropriate indicator.

6.1 Problem of the Rate of Change

In the question requirements, we need to determine whether March 2004 increase of CO2 resulted in a larger increase than observed over any previous 10-year period. Here we will deal with it separately.

We will solve this problem by determining the rate of change of CO2 concentration, which can be achieved by finding the slope of secant line in each 10-year intervals before 2004, which can be calculated by the following formula:

$$ROC = \frac{C_f - C_i}{10} \quad (6.2)$$

where C_f is the value of observed CO2 concentration level in the last year of the 10-year interval, C_i represents the value of observed CO2 concentration level in the first year of the 10-year interval. ROC denotes the rate of change.

Here we list all the 10-year time intervals before 2004 and calculate the ROC respectively:

$$\begin{aligned} ROC(1974 - 1965) &= \frac{330.19 - 320.04}{10} = 1.015ppm/year \\ ROC(1984 - 1975) &= \frac{344.87 - 331.13}{10} = 1.374ppm/year \\ ROC(1994 - 1985) &= \frac{358.96 - 346.35}{10} = 1.261ppm/year \\ ROC(2004 - 1995) &= \frac{377.70 - 360.97}{10} = 1.673PPM/year \end{aligned} \quad (6.3)$$

Since $1.673 > 1.374 > 1.261 > 1.015$, we agree with the statement that the ROC of the CO2 concentration during 2004-1995 is the largest compared with any previous time intervals.

6.2 Forecasts on Future CO2 Concentration level

In this part, we will use the regression model, ARIMA model and BP model to predict future CO2 concentration in 2050 and 2100 based on the observed CO2 concentration level data. We will also compare the results of these three model and compare the advantages as well as the disadvantages.

6.2.1 Regression Model

We will first use our linear and non-linear model to fit in the given CO2 concentration level dataset. We will evaluate the best fit expression without overfitting and the optimal degree of the regression model using a variety of indicators, such as R^2 .

We will construct piece-wise linear regression and quadratic regression (non-linear regression when the degree of the polynomial is 2) models to see whether the value of R^2 meets the criteria. ($R^2 = 0.90$). The results are shown below:

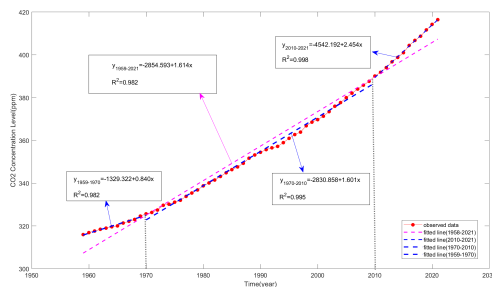


Figure 6.1: CO2 Piece-wise Linear Regression

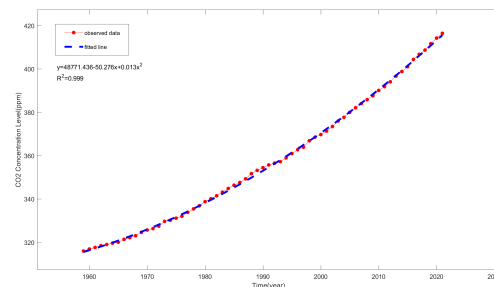


Figure 6.2: CO2 Quadratic Regression

Table 6.1: CO2 Forecasts Based on Regression Model

Year	Linear Regression Prediction	Quadratic Regression Prediction
2050	493.640ppm	485.09ppm
2100	683.858ppm	607.770ppm
R^2	0.995	0.999

In the prediction section, we use the last piece of the piece-wise linear function because the function tends to increase exponentially.

Besides, the CO2 concentration will reach 685ppm in 2134 based on the last section of linear regression model and in 2099 based on quadratic regression model.

6.2.2 ARIMA Model

As we mentioned, for CO2 concentration, we choose to use $ARIMA(3, 2, 2)$

The red dotted line represents the confidence interval of prediction (range of estimation) in Figure 6.3. We can see that the concentration level of CO2 tend to increase exponentially. The Concentration of CO2 in 2050 and 2100 predicted by ARIMA model is shown below, in Table 6.2.

In our $ARIMA(3, 2, 2)$ model, $R^2 = 0.999$, which is also a desirable outcome.

Besides, based on our model, the concentration of CO2 will reach 685 ppm in 2096.

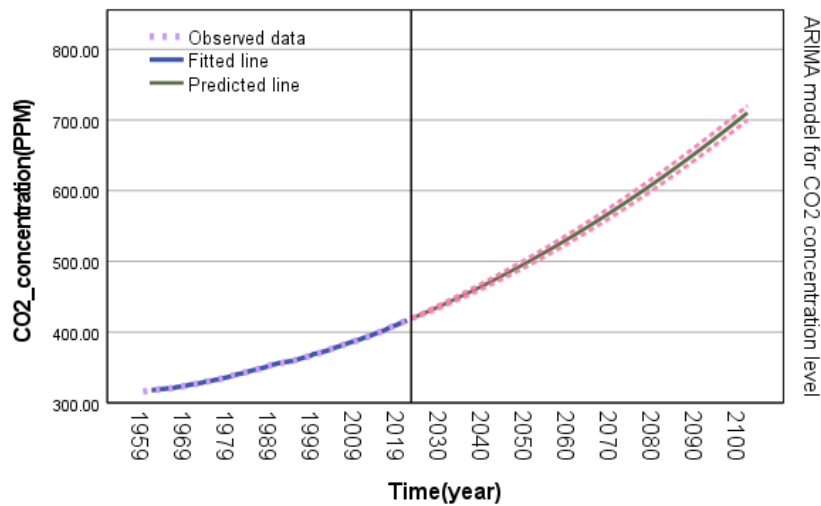


Figure 6.3: CO2 Prediction Based on ARIMA

Table 6.2: CO2 Forecasts Based on ARIMA Model

Year	ARIMA Prediction
2050	497.97ppm
2100	770.57ppm
R^2	0.999

6.2.3 BP Neural Network Model

We also use BP neural network model to forecast future CO2 concentration level. The parameter of BP Neural Network is shown as following:

Number of Input = 4, Number of Output = 1, Number of Neurons in the Hidden Layer = 9, Learning Rate = 0.01, Number of Epochs = 1000.

With the parameters given above, the prediction result is presented in Figure 6.4.

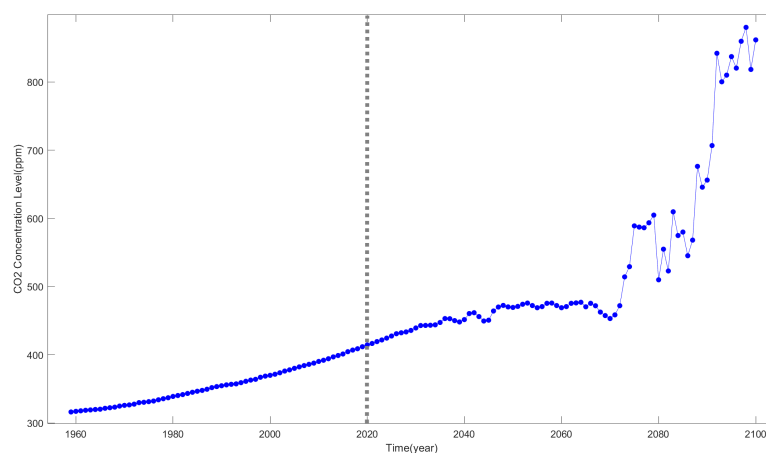


Figure 6.4: CO2 Prediction Based on BP Model

Table 6.3: CO2 Forecasts Based on BP Model

Year	BP Model Prediction
2050	469.28ppm
2100	861.73ppm
R^2	0.999

We find that the BP neural network fits extremely well ($R^2 = 0.999$) and demonstrates excellent performance in predicting changes up to 10 years. However, the overall prediction results of the neural network for CO2 were not satisfactory, demonstrating great errors in long-term predictions. We believe the reason for this is that the neural network can only predict continuously by using forecast data when predicting over long periods of time. This results in a severe reduction in accuracy.

Besides, the CO2 concentration level will reach 685ppm in 2091 based on BP neural network model.

6.2.4 Comparison Among CO2 Prediction Models

Combining the results of the models in the above three subsections, although the quadratic regression model and the ARIMA model have the same goodness of fit, we believe that the most suitable model for forecasting the CO2 trend is the ARIMA model, because ARIMA model is the combination of several small models and it also takes seasonal factors as well as noises into account, which is more comprehensive than the regression model.

Although the BP model fits extremely well and shows accuracy in short-term forecasting, we do not believe that the BP model can accurately predict the long-term future.

6.3 Forecasts on Future GMSAT

In this part, we will use the regression model, ARIMA model and BP model to predict future GMSAT based on the observed data. We will also compare the results of these three model and compare the advantages as well as disadvantages.

6.3.1 Regression Model

We first use our linear and non-linear (quadratic) regression model to fit the given GMSAT dataset, and see whether the criteria is met.

We summarize our prediction in the following table:

Table 6.4: GMSAT Forecasts Based on Regression Model

Increase of GMSAT	Linear Regression Prediction	Quadratic Regression Prediction
1.25 °C	2044	2031
1.50 °C	2060	2038
2.00 °C	2090	2052
R^2	0.891	0.918

The R^2 of the quadratic regression ($R^2 = 0.918$) meets the criteria ($R^2 = 9$), and it is

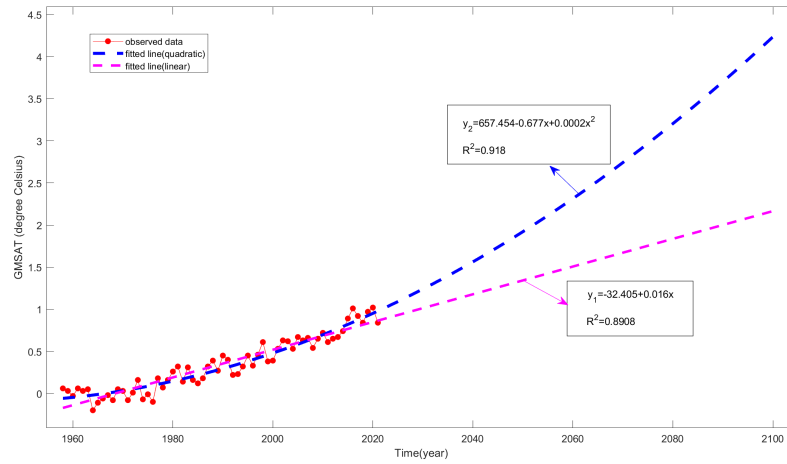


Figure 6.5: GMSAT Linear and Quadratic Regression Models

also slightly higher than the R^2 of the linear regression ($R^2 = 0.891$). Hence, in this case, we consider quadratic regression as the best fit regression model.

6.3.2 ARIMA Model

In 4.3 Results, we discovered that the best parameter for GMSAT ARIMA model is $ARIMA(3, 1, 2)$. Here we present our results:

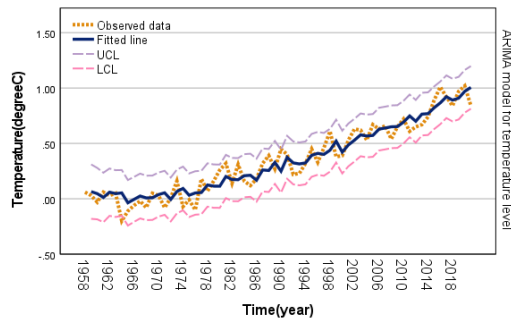


Figure 6.6: GMSAT ARIMA Model

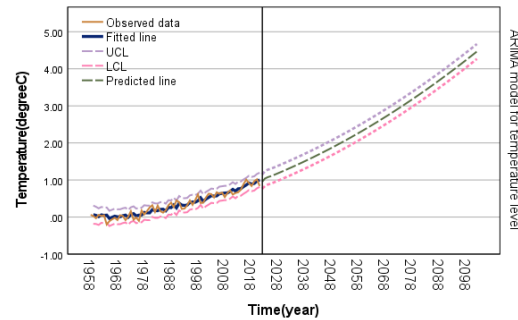


Figure 6.7: GMSAT ARIMA Model Prediction

Table 6.5: GMSAT Forecasts Based on ARIMA

Increase of GMSAT	ARIMA Model Prediction
1.25 °C	2030
1.50 °C	2038
2.00 °C	2052
R^2	0.918

6.3.3 BP Network Model

We also use BP network to forecast future GMSAT. Our neural network is based on the same parameters mentioned in 6.2.3. The following graph is the prediction results:

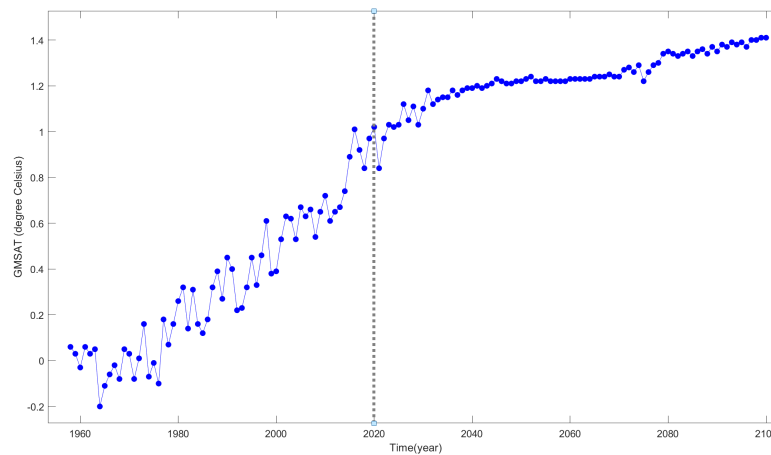


Figure 6.8: GMSAT Prediction Based on BP Model

Table 6.6: GMSAT Forecasts Based on BP Neural Network Model

Increase of GMSAT	BP Model Prediction
1.25 °C	2068
1.50 °C	unknown
2.00 °C	unknown
R^2	0.948

We can see that in this scenario, BP model still displays accuracy ($R^2 = 0.948$) in the prediction of the first 10 values. However, the data for GMSAT are in a state of almost stagnant growth for long-term projections, which are not reliable somehow.

6.3.4 Comparison Among GMSAT Prediction Models

Combining the information from the first 3 subsections, we conclude that both ARIMA model and quadratic regression model are well suited in predicting future GMSAT, for that both models are almost exactly similar in both predicted values and goodness-of-fit.

BP neural network model is still excellent in predicting short-term GMSAT but lacks the capacity in long-term forecasts.

7 Correlation Model of CO₂ and Temp

7.1 General Approach

Correlation Coefficient is a very commonly used statistical measure that usually describes the strength of the relationship between two variables in terms of a specific coefficient. The main purpose of the correlation model is to measure the potential correlation between CO₂ concentrations and land-ocean temperatures by calculating the correlation coefficients.

7.1.1 Interpretation of Pearson correlation coefficient

The first correlation coefficient we used to determine whether there is a linear relationship between CO₂ concentration and land-ocean temperature is the Pearson correlation coef-

ficient(PCC). Before getting deeper, we first introduce some basic statistical concepts that we will use in our calculation of PCC:

$$cov_{(C,T)} = \frac{\sum_{i=1}^n (C_i - \bar{C})(y_i - \bar{T})}{n - 1} \quad (7.1)$$

$$s_C = \sqrt{\frac{\sum_{i=1}^n (C_i - \bar{C})^2}{n - 1}} \quad , \quad s_T = \sqrt{\frac{\sum_{i=1}^n (T_i - \bar{T})^2}{n - 1}} \quad (7.2)$$

$$\bar{C} = \frac{1}{n} \cdot \left(\sum_{i=1}^n C_i \right) \quad , \quad \bar{T} = \frac{1}{n} \cdot \left(\sum_{i=1}^n T_i \right) \quad (7.3)$$

where n represents the size of our samples (CO2 and temperature).

Here we briefly explain the meaning of those statistic terms.

- The covariance (7.1) is used to measure the overall error between CO2 concentration level and temperature and is used to reflect the relationship between them.
- The standard deviation (7.2) is the arithmetic square root of the variance and is used to reflect the degree of dispersion.
- The mean (7.3) is used to find the mathematical average of CO2 concentration level or temperature.

We calculate the value of PCC by using sample covariance to divide the product of sample standard deviation, so the influence on covariance triggered by the rate of change of variables in that covariance can be eliminated. To some extent, the value of Pearson coefficient will reflect the two variables temperature and concentration when they fluctuate per unit:

$$r_p = \frac{cov_{(C,T)}}{S_C \cdot S_T} = \frac{\sum_{i=1}^n (C_i - \bar{C}) \cdot (T_i - \bar{T})}{\sqrt{\sum_{i=1}^n (C_i - \bar{C})^2} \cdot \sqrt{\sum_{i=1}^n (T_i - \bar{T})^2}} \quad (7.4)$$

where r_p is PCC

r_p takes values in the range $[-1,1]$, so we will evaluate the correlation by the absolute value of it ($|r_p|$). Usually, the linear correlation strength of CO2 concentration level and land-ocean temperature can be judged by the following range of values:

Table 7.1:	
$ r_p $	Correlation Strength
0.8-1.0	Strong Correlation
0.6-0.8	Relatively Strong Correlation
0.4-0.6	Medium degree of Correlation
0.2-0.4	Relatively Weak Correlation
0.0-0.2	Weak or No Correlation

7.1.2 Interpretation of Spearman's rank correlation coefficient

To help us acquire a profounder understanding about the correlation between CO2 and land-ocean temperature. We introduce another concept of correlation coefficient - Spearman's rank correlation coefficient (SRCC), which is used to determine the strength of the monotonic relationship between CO2 and land-ocean temperature.

We first need to rearrange the sample values of the two random variables (CO2 concentration level and land-ocean temperature) in order of the magnitude of the variables, and then use the order of the sample values instead of the actual data.

SRCC can be determined as following:

$$r_s = r_{R(C)R(T)} = \frac{\text{cov}(R(C), R(T))}{S_{R(C)} \cdot S_{R(T)}} = \frac{\sum_{i=1}^n (R(C_i) - \overline{R(C)}) \cdot (R(T_i) - \overline{R(T)})}{\sqrt{\sum_{i=1}^n (R(C_i) - \overline{R(C)})^2} \cdot \sqrt{\sum_{i=1}^n (R(T_i) - \overline{R(T)})^2}} \quad (7.5)$$

where $R(C)$ and $R(T)$ represents rank variables, which refers to the order of variables in a numerical sequence. Cov and S represents covariance (7.1) and standard deviation (7.2) respectively.

SRCC indicates the direction of correlation between CO2 concentration and land-ocean temperature. We will interpret R_s from several aspects:

- A positive coefficient r_s represents a monotonically increasing relationship. On the opposite, a negative coefficient represents a monotonically decreasing relationship.
- $r_s = 0$ indicates that there is no tendency between the CO2 and land-ocean temperature.
- When C and T are perfectly monotonically correlated, the absolute value of the SRCC is 1. The monotonicity is stronger when the absolute value is approaching to 1.

7.2 Solving For PCC and SRCC

We calculate the PPC and SRCC between CO2 concentration level and land-ocean temperature based on the given dataset. The results are shown below:

$$\begin{aligned} r_p &= 0.961 \\ r_s &= 0.949 \end{aligned} \quad (7.6)$$

Because the value of $|r_p|$ is between 0.8 and 1, there is a strong linear correlation between CO2 concentration level and land-ocean temperature (table 7.1), and because the value of $|r_s|$ is positive and it is closed to 1, we affirm that there is also a strong monotonic increasing relationship between them.

7.3 Construction of the Correlation Model

Considering the results we got in section 7.3, we decided to construct a linear regression model to describe the relationship between them.

First, we determine the expression of our correlation model:

$$T = aC + b \quad (7.7)$$

We determine the best fit coefficients a and b by LSM, which we mentioned in section 3.2.

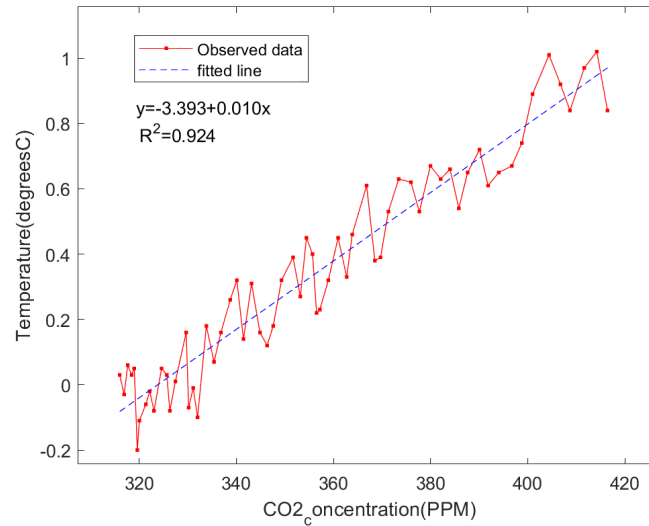


Figure 7.1: Linear Regression Model between CO2 and Temp

By constructing a simple linear regression model, we can get the optimize coefficients:

$$T = 0.010C + 3.393 \quad (7.8)$$

$$a = 0.010 \quad b = 3.393 \quad R^2 = 0.924$$

To maximize the precision and increase the value of R^2 , we continue to use the ARIMA model to fit the residual term (the method we mentioned in 4. extended regression model). We use ACF and PACF to determine the best parameters for the ARIMA model(q,d,q).

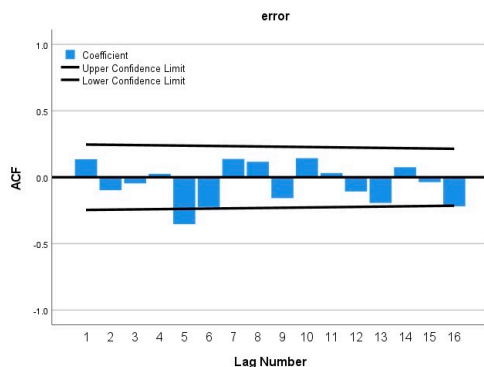


Figure 7.2: ACF of Residual Terms

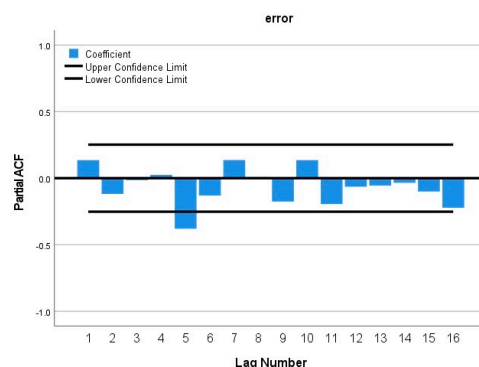


Figure 7.3: PACF of Residual Terms

This is the original ACF and PACF diagram of the residual term. Because there is no geometric decay, we do not need to difference the initial residual value. Therefore $d=0$. Also, we

find that in the ACF diagram, the lag cuts off after $p=3$ and in the PACF diagram, the lag cuts off after $q=2$. Hence, we construct $ARIMA(2,0,3)$ model to fit in the residual terms.

We therefore construct a of multivariate linear regression model by considering $ARIMA(2,0,3)$ and C as two variables. Ultimately, we obtain the CO_2 temperature correlation model as:

$$T = 3.432 + 0.011C + 1.041 \cdot ARIMA(2, 0, 3) + \varepsilon$$

$$\beta_0 = 3.432 \quad \beta_1 = 0.011 \quad \beta_2 = 1.041 \quad R^2 = 0.936 \quad (7.9)$$

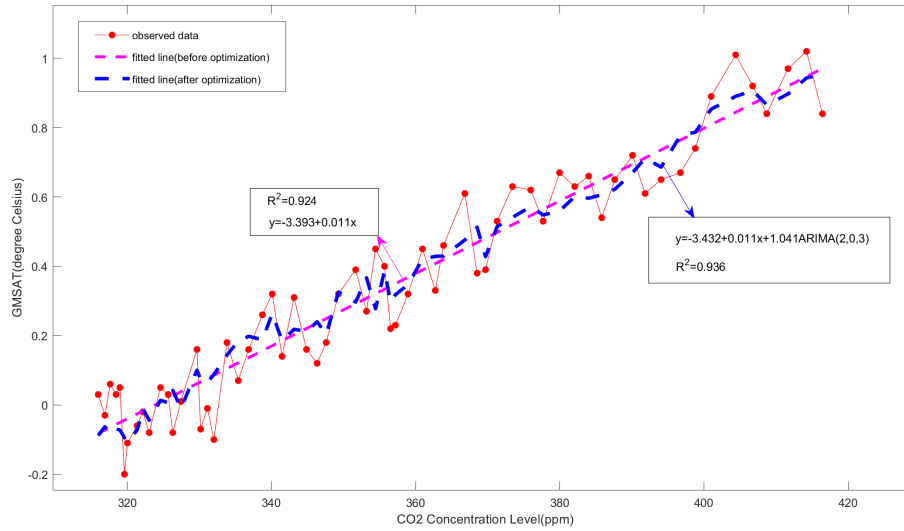


Figure 7.4: Extended Linear Regression Model Based on ARIMA

The value of R^2 increases 0.012 by adding ARIMA model and taking noises into account. Hence, the goodness-of-fit could be maximized. (Figure 7.4)

7.4 Reliability of the Correlation Model

We test the reliability of the correlation model using CO_2 ARIMA model as well as the quadratic regression model for temperature. We performed a linear regression and use CO_2 future values (predicted by ARIMA) and temperature values (predicted by quadratic regression) as two variables, comparing the best fit coefficients with the parameters of the current correlation model. The graph is shown in Figure 7.5.

Based on Figure 7.5, we find that the two lines largely coincide when CO_2 is less than 470 ppm (about 2040), and we can conclude that the correlation model is largely reliable until 2040. As time passes, the error increases, and at about 650 ppm CO_2 (about 2090), the percentage error reaches 20%, increasing at an average rate of 3% per year. We use the formula below to calculate the percentage error:

$$PercentageError = \frac{|v_a - v_e|}{v_e} \cdot 100\% \quad (7.10)$$

where v_a is the actual and v_e represents expected value.

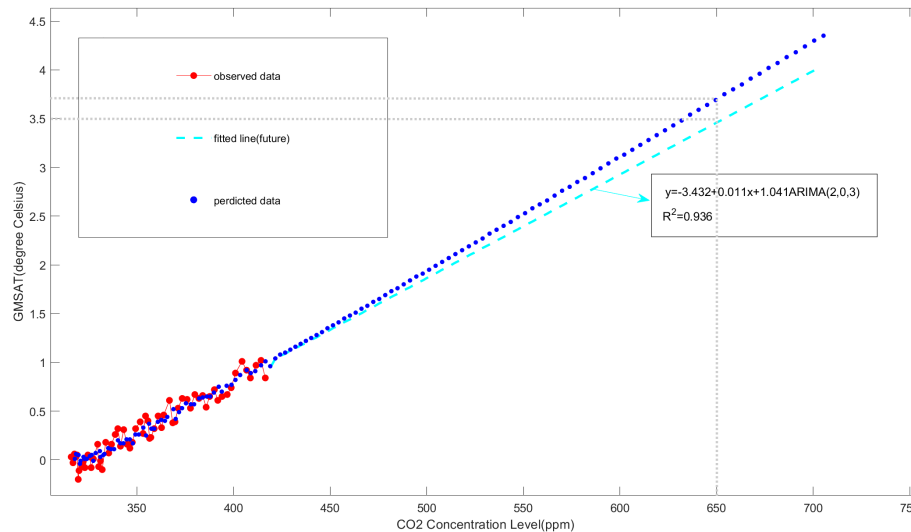


Figure 7.5: Reliability of the Correlation Model

8 Sensitivity Analysis

8.1 BP Model

- **Number of Input** We find that when the number of inputs < 4 , the output CO2 concentration or GMSAT will fluctuate significantly. The range of output will become large and unpredictable, which corresponds to the result we expected. When the number of inputs > 4 , the value of output will not change significantly.
- **Number of Output** When the output is greater than three, the last few outputs will always produce a large error.
- **Number of Neurons** We find that some small error will be produced after the number of neurons < 4 , the error is around 2-4. When $9 < \text{number of neurons} < 100$, the value of the output value does not change significantly. However, as the number of neurons goes up, the training time also increases.
- **Number of Epochs** For CO2 concentration level prediction, it is surprising that if the number of epochs is more than 10, R^2 could always reach 0.99. When it comes to GMSAT, the number of epochs must exceed 500 to guarantee a acceptable R^2 .

8.2 ARIMA Model

- **CO2 ARIMA Model** We conducted sensitivity tests for each of the three parameters. However, the value of R^2 only changes subtly. We believe this could indicate that the fluctuation of the CO2 concentration time series is quite regular.
- **GMSAT ARIMA Model** After we change the value of p, d, and q, the R^2 of the function decreases slightly. When we change the value of q by 1, the largest decrease happens (around 0.2).

9 Strengths and Weaknesses

9.1 Strengths

- **Multiple Models** To predict future changes in CO₂ and temperature, we have developed a variety of models, each based on different methods and theories. We compared the goodness-of-fit of each model horizontally and selected the best models for CO₂ and temperature forecasts. Thus, the strength is that the prediction results for the land-ocean temperature and CO₂ concentration level are very comprehensive, because our considerations are based on multiple models.
- **The goodness-of-fit is high** The goodness of fit is high in the regression, ARIMA, BP and correlation models, proving that all the models we have built fit the provided data set very well
- **Models are excellent in both short and long term forecasting** BP models are well suited to predict short-term climate change because of their excellence in the goodness-of-fit. Regression and ARIMA models, on the other hand, are well suited for predicting long-term climate change by considering the trend of the entire data set.
- **Combining Multiple Models** We construct extended a linear regression model, which is the combination of ARIMA and linear regression. By utilizing such method, the goodness-of-fit gets improved.

9.2 Weaknesses

- **Assumption is somehow strong** Because there are a lot of uncertainties on climate, our predictive model is only used to anticipate future trend based on historical data. We cannot predict the future situation in which the climate is extremely affected by extreme factors.
- **BP model has room for improvement in long-term forecasting** Because the data is limited, neural network cannot study and analysis the data very well.

10 Conclusion and Future Work

We have developed three different mathematical models based on different theories and successfully completed future predictions of CO₂ and temperature, as well as successfully simulated the relationship between CO₂ and temperature. At present, using of BP model enables us to forecast short-term data accurately. Nevertheless, with the dearth of dataset, long-term forecasts are not reliable to some extent. Therefore, our future work is to try to optimize this model by collecting more data and creating a larger training matrix for neural network.

References

- [1] Deepai, BackPropagation <https://deepai.org/machine-learning-glossary-and-terms/backpropagation>
- [2] The Kolmogorov–Arnold representation theorem revisited, Johannes Schmidt-Hieber, University of Twente and Leiden University, Drienerlolaan 5, 7522 NB Enschede, The Netherlands

Feeling the Heat:

Global Warming and CO₂ Emission

"4.3 million acres of land — an area the size of Connecticut — will be inundated due to sea level rise by 2050??!" [1] This is a precise estimate of the pace of sea level rise made by scientists, not a rumor. By 2050, sea level rise could result in over \$35 billion in economic losses to real estate. Therefore, the study of global warming and how to effectively curb it has become a focal point of discussion in the global scientific community.

According to NOAA's Global Climate Report 2021, global temperatures rise by 0.84 degrees Celsius in 2021 compared to the mean. According to our rigorous and thorough assessment models, future warming will continue, rising by 1.25 degrees Celsius in 2031, 15 degrees Celsius in 2038 and 2 degrees Celsius in 2052. In 3300, according to the current development trend, the Earth's temperature will increase by 20 degrees Celsius, and the summer temperature will reach 60 degrees Celsius in some areas, which will cause extreme damage to the global ecosystem and make the Earth no longer suitable for human habitation. Meanwhile, in order to explore the factors affecting global warming, our team used mathematical methods, focusing on analyzing and processing CO₂ emission data as well as predicting the future emission of CO₂. According to our mathematical model, CO₂ will reach 498 ppm by 2050 and will reach 770 ppm by 2100.



[3]



[2]

Ultimately, we found there is a strong linear relationship between CO₂ concentration and land-ocean temperature, when CO₂ concentration level increases, temperature rises correspondingly. More specifically, our mathematical model predicts that a doubling of atmospheric CO₂ would result in a 4 to 4.5 degree Celsius rise in global temperature. Based on our predicted data, the current situation of earth is not as optimistic as before, the global temperature raises dramatically without any control. Carbon dioxide emissions can be deemed as the primary driver of global climate change. We suggest that to avoid the worst impacts of climate change, it is needed to urgently reduce CO₂ emissions. Under this circumstance, individuals are responsible to reduce their carbon footprints by avoiding buying fast fashion, considering purchasing a hybrid or electric vehicle, or changing incandescent light bulbs.

[1] <https://www.scientificamerican.com/article/35-billion-worth-of-real-estate-could-be-underwater-by-2050/> Scientific American

[2] <https://www.cinconoticias.com/humano-devastadora-pandemia-del-planeta/>

[3] <https://www.artstation.com/artwork/3oQAKB>